Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A survey was conducted on students' preferred modes of transportation to school. The results are summarized in the two-way frequency table below. \\ | | Bus | Bicycle | Walking | Total | |----------------|-----|---------|---------|-------| | Under 16 | 45 | 30 | 25 | 100 | | 16 and Over | 60 | 15 | 25 | 100 | | Total | 105 | 45 | 50 | 200 | Let B be the event that a randomly selected student prefers to travel by Bus. Let U be the event that a randomly selected student is Under 16 years old. \\ What is the probability that a student prefers to travel by Bus, given that they are Under 16 years old? Mathematically, this is denoted as $P(B|U)$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A civil engineer measures the width of a river BC by setting up parallel stakes such that $DE \parallel BC$. Point D is on line AB and E is on line AC. If $AD = 45$ m, $DB = 75$ m, and $DE = 36$ m, what is the width of the river BC?

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A 5.5-foot-tall surveyor casts a 4.4-foot shadow on level ground. At the exact same moment, a nearby radio antenna tower casts a 36-foot shadow. How tall is the antenna tower?

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC, segment $DE \parallel BC$. The segment lengths are $AD = x$, $DB = x + 3$, $AE = x + 2$, and $EC = x + 7$. What is the value of x?

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC, ray AD bisects $\angle BAC$ with D lying on side BC. If $AB = x + 3$, $AC = 2x$, $BD = 4$, and $DC = 6$, what is the value of x?

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In an hourglass figure where $AB \parallel CD$ with intersection at E, $AE = 2x + 1$, $EC = 5x - 4$, $BE = 6$, and $ED = 9$. What is the value of x?

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In right ΔABC with altitude $CD \perp AB$, $AD = 2$ and $DB = 8$. What is the ratio of the legs $\frac{AC}{BC}$?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two similar triangles have areas of $45\ \text{cm}^2$ and $80\ \text{cm}^2$. If the perimeter of the smaller triangle is $27\ \text{cm}$, what is the perimeter of the larger triangle?

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

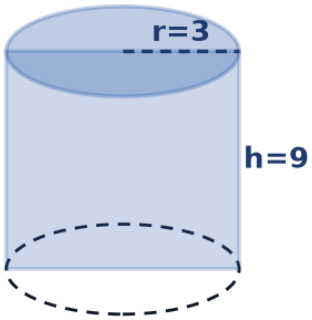
The point P has coordinates $(-3, 4)$. Point P' is the image of P after a reflection across the line $y = x$. What are the coordinates of P'?

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In right ΔABC with right angle at C and altitude $CD \perp AB$, leg $BC = 12$ and hypotenuse $AB = 16$. What is the length of segment DB?

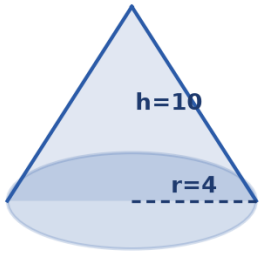
11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 5x + 8$ in volume of prisms and cylinders, evaluate $f(5)$.



12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 9$ in volume of pyramids and cones, evaluate $f(2)$.



13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for volume of pyramids and cones consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · AdvancedIn the hard Module 2, questions 18–22 are typically high-difficulty multi-step problems with parameters k. If a student has 7 minutes remaining for the last 3 questions, which approach optimizes score yield: attempting all 3 partially, or using Desmos regression and choice elimination to guarantee complete accuracy on 2 of them?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A circular archery target has three concentric rings of radii 10 cm, 20 cm, and 30 cm. If an arrow hits the target completely at random, what is the exact probability that it lands in the middle ring (between radii 10 cm and 20 cm)?

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Circles

A circle is inscribed inside a square of side length 10 cm. A point is chosen at random inside the square. What is the exact probability that the point lands outside the circle (in the four shaded corners)?

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Area, Surface Area & Volume

[Probability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Transformations & Symmetry

A cone and a cylinder have identical base radii of 6 cm. If both solids have the same volume of $180\pi\text{ cm}^3$, find the height of the cylinder and the height of the cone.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Foundations & Reasoning

A rhombus has diagonals of lengths 16 cm and 12 cm. Find the perimeter and area of the rhombus.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Coordinate Geometry

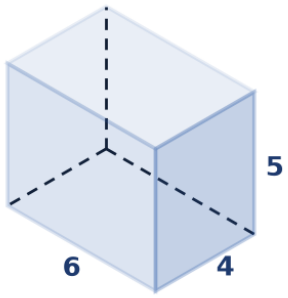
From an external point P, two tangents PA and PB are drawn to a circle with radius $r = 5\text{ cm}$. The distance from P to the center of the circle O is 13 cm. Find the length of the tangents PA and PB, and find the perimeter of quadrilateral $PAOB$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 6$ in cross sections of solids, evaluate $f(2)$.



a. 12

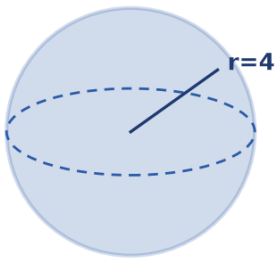
b. 15

c. 6

d. 16

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 7$ in solids of revolution, evaluate $f(2)$.



a. 15

b. 19

c. 8

d. 19

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for cross sections of solids consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solids of revolution consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for volume of prisms and cylinders consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for volume of spheres and hemispheres consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 3$ in volume of spheres and hemispheres, evaluate $f(3)$.

a. 9

b. 11

c. 6

d. 13

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for volume of composite solids consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 3x + 4$ in volume of composite solids, evaluate $f(3)$.

a. 13

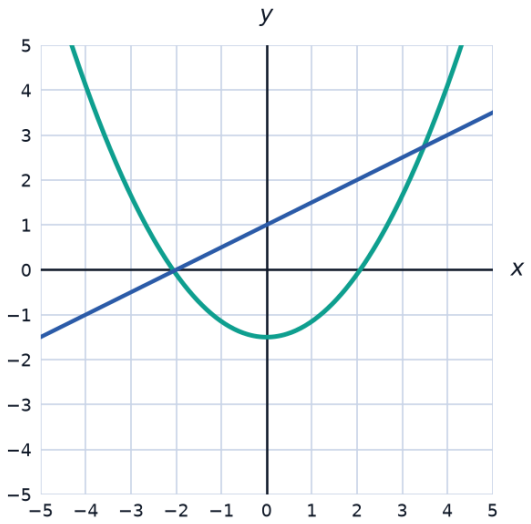
b. 16

c. 9

d. 17

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A point is chosen at random on a segment of length 11 cm. Find the probability it lands within 2 cm of the left endpoint.



a. $\frac{1}{6}$

b. $\frac{2}{11}$

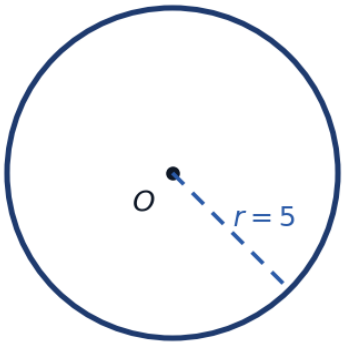
c. 1

d. $\frac{9}{11}$

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31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A circular dartboard has radius 15 cm with a bullseye of radius 1 cm at the center. Assuming a dart lands at a uniformly random point on the board, find the probability it lands in the bullseye.



a. $\frac{1}{226}$

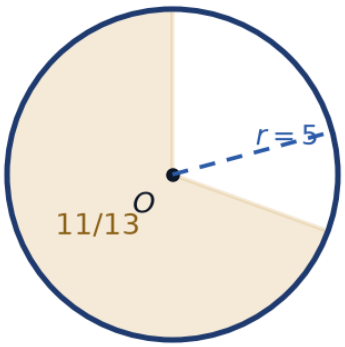
b. $\frac{1}{225}$

c. $\frac{224}{225}$

d. $\frac{1}{15}$

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

The probability that an event occurs is $\frac{11}{13}$. What is the probability that the event does NOT occur?



a. 0

b. $\frac{2}{13}$

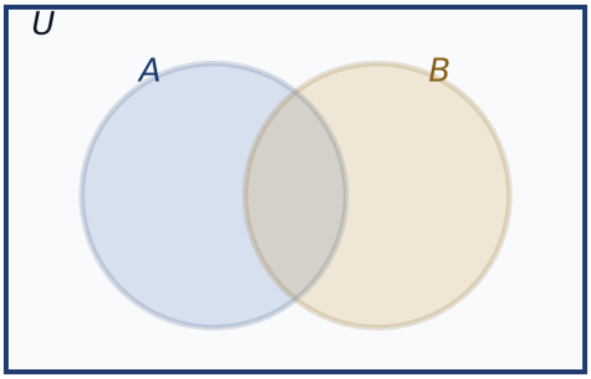
c. 1

d. $\frac{11}{13}$

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Independent events A and B have $P(A) = \frac{1}{5}$ and $P(B) = \frac{2}{5}$. Find $P(A \text{ or } B)$.



a. $\frac{3}{5}$

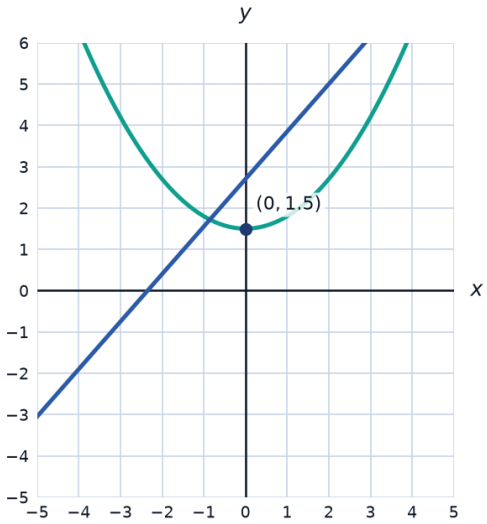
b. $\frac{14}{25}$

c. $\frac{2}{25}$

d. $\frac{13}{25}$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A circular target of radius 11 cm has a ring-shaped region between radius 3 cm and radius 9 cm. Find the probability a randomly landing dart hits this ring.



a. $\frac{9}{121}$

b. $\frac{81}{121}$

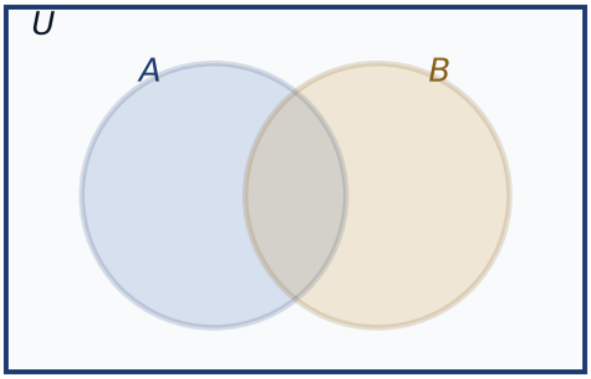
c. $\frac{72}{121}$

d. $\frac{36}{61}$

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two independent events A and B have $P(A) = \frac{1}{5}$ and $P(B) = \frac{6}{12}$. Find $P(A \text{ and } B)$.



a. $\frac{1}{10}$

b. $\frac{2}{11}$

c. $\frac{1}{5}$

d. $\frac{1}{2}$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A quality control laboratory receives components sorted into two bins: - **Bin A** contains 10 components, of which exactly 3 are defective and 7 are non-defective. - **Bin B** contains 10 components, of which exactly 6 are defective and 4 are non-defective. To conduct an inspection, an automated system first selects a bin at random: it rolls a fair standard six-sided die and selects **Bin A** if the result is 1 or 2; otherwise (if the result is 3, 4, 5, or 6), it selects **Bin B**. From the chosen bin, the system then randomly samples 2 components without replacement. If both sampled components are found to be defective, what is the probability that the components were drawn from **Bin A**?

a. $\frac{1}{11}$

b. $\frac{1}{9}$

c. $\frac{1}{6}$

d. $\frac{10}{11}$

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A random sample of 200 high school students was surveyed regarding their grade level and whether they participate in school sports. The results are summarized in the two-way frequency table below: | | Participates in Sports (S) | Does Not Participate in Sports | Total | | :--- | :---: | :---: | :---: | | Grade 10 (T) | 48 | 32 | 80 | | Grade 11 | 42 | 78 | 120 | | Total | 90 | 110 | 200 | A student is selected at random from this sample. Let S be the event that the student participates in sports, and let T be the event that the student is in Grade 10. Which of the following statements is correct regarding the conditional probability $P(S \mid T)$ and the statistical independence of events S and T?

a. $P(S \mid T) = 0.60$, and the events are not independent because $P(S \mid T) eq P(S)$.

b. $P(S \mid T) = 0.60$, and the events are independent because $P(S \mid T) > P(S)$.

c. $P(S \mid T) = 0.24$, and the events are not independent because $P(S \mid T) eq P(S)$.

d. $P(S \mid T) = 0.40$, and the events are independent because $P(S \mid T) = P(T)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Transformations & Symmetry

A bag contains 5 red marbles and 4 blue marbles. Two marbles are drawn from the bag without replacement. What is the probability that the first marble drawn is red and the second marble drawn is blue?

a. $\frac{5}{18}$

b. $\frac{4}{18}$

c. $\frac{20}{81}$

d. $\frac{5}{9}$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Foundations & Reasoning

A point is chosen uniformly at random on the line segment from $x = 0$ to $x = 20$. Let event A be the event that the point lands on the interval $[2, 10]$, and let event B be the event that the point lands on the interval $[6, 16]$. What is the probability that the point lands in A or B, $P(A \cup B)$?

a. $\frac{7}{10}$

b. $\frac{9}{10}$

c. $\frac{1}{5}$

d. $\frac{1}{2}$

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Coordinate Geometry

A bag contains 18 marbles, of which 12 are red. If one marble is drawn at random, what is the probability it is red?

a. $\frac{1}{3}$

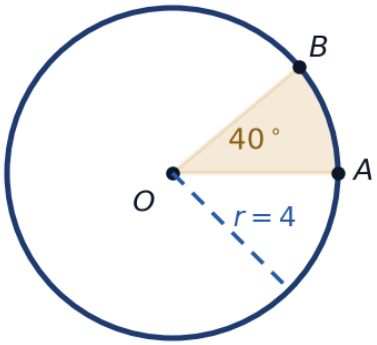
b. $\frac{2}{3}$

c. $\frac{3}{2}$

d. $\frac{12}{19}$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Lines & Angles

Two secants are drawn from an external point to a circle, intercepting arcs of 100° and 40°. What is the measure of the angle formed by the secants?



a. 100°

b. 30°

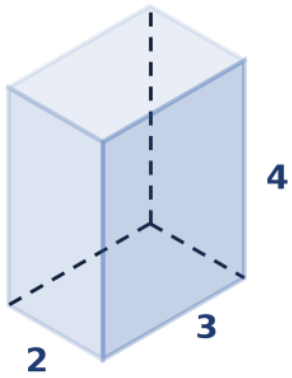
c. 70°

d. 40°

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Triangles & Congruence

Volume of a rectangular prism with dimensions $2 \times 3 \times 4$:



a. 21

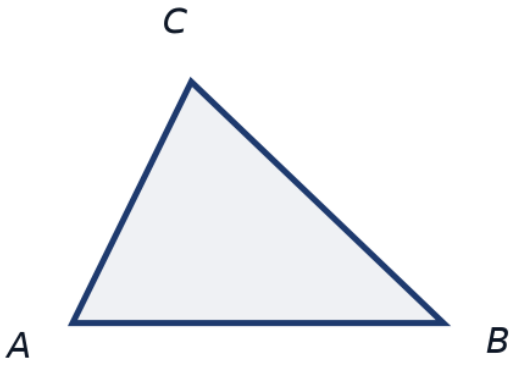
b. 9

c. 24

d. 26

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

In a triangle, the two remote interior angles measure 14° and 10°. What is the measure of the exterior angle at the third vertex?



a. 24°

b. 14°

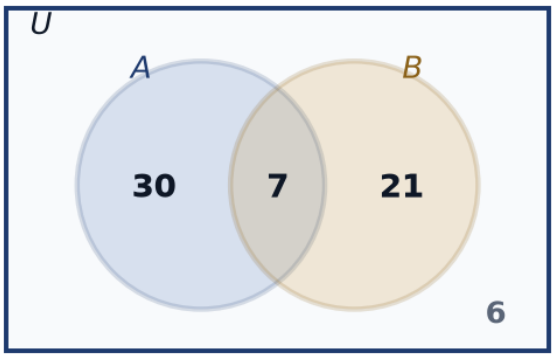
c. 156°

d. 10°

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

Two chords intersect inside a circle, intercepting arcs of 30° and 178° (the arc and its vertical arc). What is the measure of the angle formed at the intersection?



a. 76°

b. 208°

c. 30°

d. 104°

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A rectangle has length 11 and width 6. What is its area?



a. 34

b. 66

c. 72

d. 17

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